Is there anybody out there?

From the dark side of the moon to the eternal vastness of the universe, are we on the verge of contact?

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On the night of August 15, 1977, the Big Ear telescope – located at Ohio State University, USA – received a strong radio signal from the Sagittarius constellation, which would eventually create radio astronomy history. Constructed in 1961 and functional since 1963, the Big Ear radio telescope was part of the Search for Extra Terrestrial Intelligence (SETI) project. Since the project began in 1973, it ended up being one of the longest running SETI projects ever, until it was shutdown in 1995. Astronomer Jerry Ehman who had volunteered in analysing the large amounts of data – collected from extragalactic radio sources – was in for a surprise while he was studying the data on printed perforated paper. The radio data was so startling for him that while encircling the anomalous data points, he commented "Wow!" on the side. This is how it popularly came to be known as the "Wow! Signal".

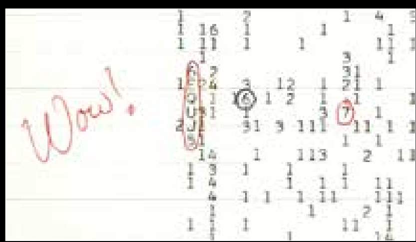
Fast forward to the present time, and we've received a lot of such mysterious signals from the infinite corners of the cosmos. Although we are yet to witness something as strong and spectacular as the Wow! Signal, the source remains a mystery to this date. Astronomers have been able to roughly pinpoint the galaxies and constellations from where these unknown signals have originated. But we haven't found success in decrypting these signals, if at all we consider it with high hopes that these mysterious signals are actually attempts to make contact by extra-terrestrials. Or were they mere radio bursts emitted from exploding stars or other celestial bodies? That's what scientists and astronomers are trying to review.

But what are these signals? How are they different from the ones being transmitted across the planet? Who takes responsibility of reviewing through the endless number of signals received by our telescopes and satellites? We try to delve deeper into these questions.

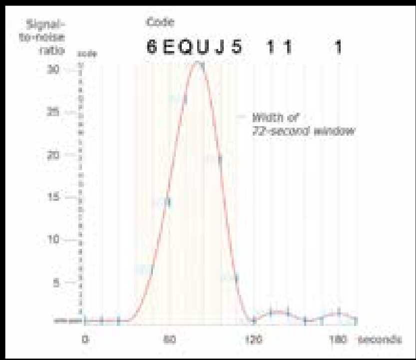
All about space signals

We've been using radio waves operating at different frequencies primarily as a channel for communication within the planet and beyond. When it comes to interstellar communication, we've long known that the electromagnetic wave the best medium to adopt because of its resistance to any form of interference and dispersion. Since this is the universally accepted form of communication we already use, our best opportunity to establish contact with any extra-terrestrial source had to be accomplished using radio waves.

Emitting radio signals at unknown frequencies will definitely go in vain hence, a favourable frequency region had to be decided. A paper titled "Searching for Interstellar Communications" published in 1959 by Cornell physicists Philip Morrison and Giuseppe



The Wow! Signal sequence



The plot of the Wow! Signal's intensity versus time